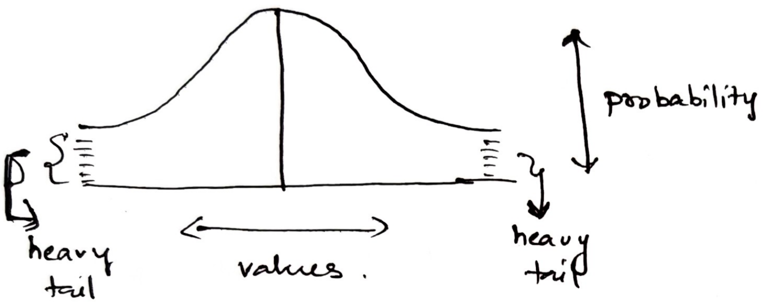
Extras (~~the Ricky Gervais show~~) (on hypothesis testing)

- Avinash

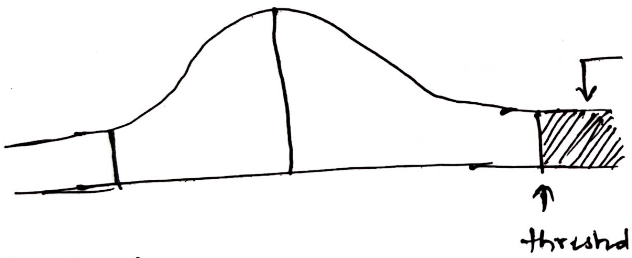
- What is t-distribution?
  * It is a probability distribution, used in inferential statistics
  * Used when sample size is small, or S.D. is unknown



Similar to normal distribution, but with heavier tails

higher chance ← for value to be farther from mean.

- What is threshold value?
  * Critical t-value (or z-value in z-distribution) that determines the boundary of rejection in hypothesis testing.
  *



probabilities here are rare, so whatever happening here is significant (or in other words, significantly diff. from mean)

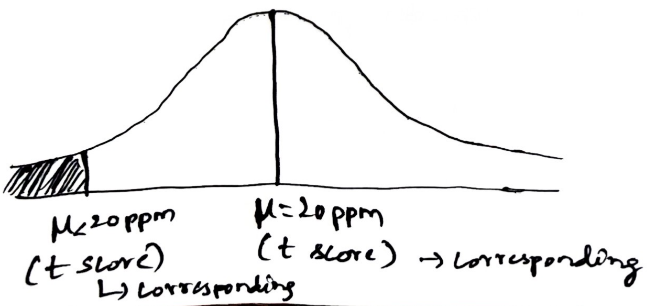
↳ can reject H_0

  * threshold value
    ↳ obtained from t-tables/z-tables

- Types of t-tests

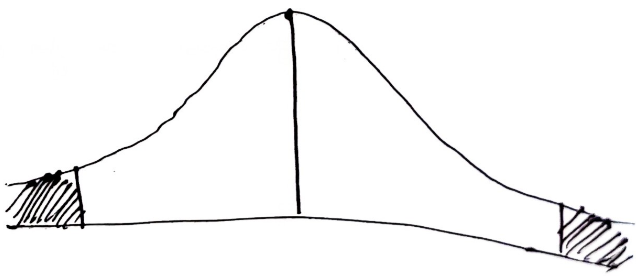
One-tailed

Used when checking against mean in one specific direction



Two-tailed

Used when checking for significant difference in either direction



- what are degrees of freedom?

* It represents how many values in a dataset are free to vary when estimating a statistical parameter

- consider 3 numbers whose mean is 10

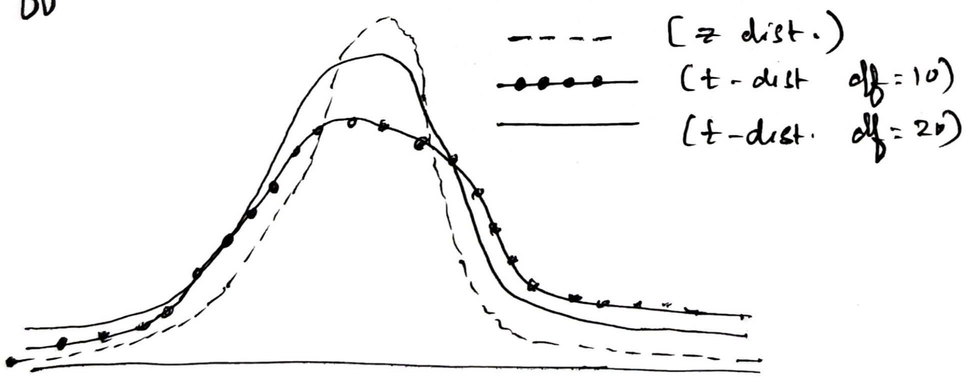
$$\therefore \frac{x_1 + x_2 + x_3}{3} = 10$$

$$\Rightarrow x_1 + x_2 + x_3 = 30$$

you are free to choose values of x_1 and x_2 (any 2 no.'s) but ~~you~~ they will dictate x_3 value, since condition is sum = 30.

$$\therefore df = n - 1 \quad \text{(where n is sample size)}$$

- Diff. b/w t-distribution & z-distribution.

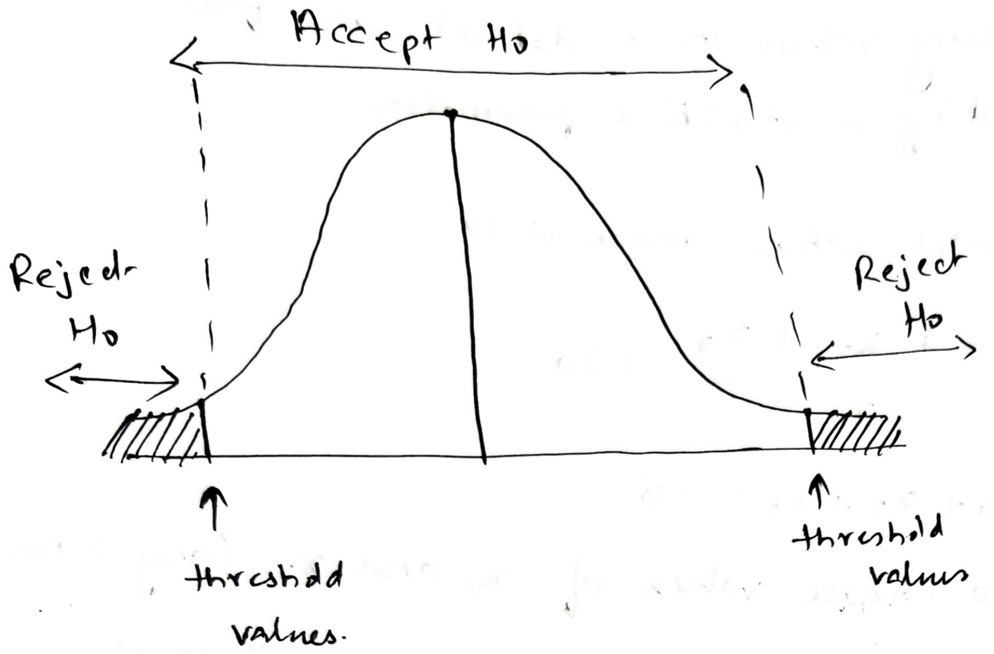


z-distribution → Larger sample, therefore less variation, more bell curve

→ mean : Avg.

→ S.D. : measure of variation of values of a variable about its mean.

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$


Accept Ho
Reject Ho
Reject Ho
threshold values.
threshold values